Supplementary Material

When Simple Is Enough: Leakage-Free Benchmarking of Classical Machine Learning vs. Graph Neural Networks in Low-Data Bioactivity Prediction

This document holds the material that supports the manuscript without being part of its argument: the architecture chosen by the inner loop in every outer fold, and the detail of the screening, docking and ADMET stage, whose results are hypotheses for experimental testing rather than conclusions of the benchmark. Every value is generated by the analysis pipeline and read from the same files as the manuscript.

Benchmark detail

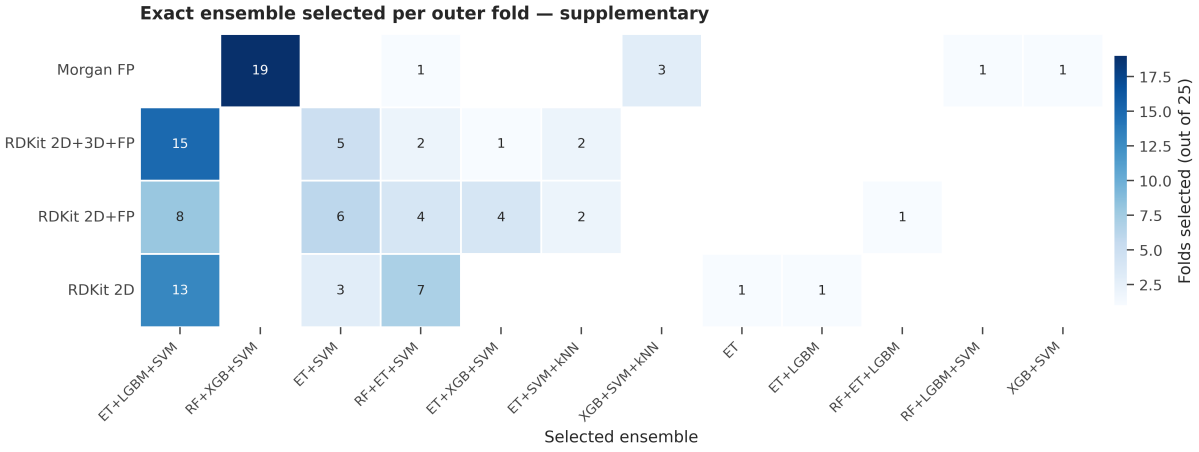


Figure S1: Stacking ensemble selected by the inner loop in each of the 25 outer folds, for each molecular representation. Cell values are the number of outer folds in which that exact combination of base learners (RF, random forest; ET, extra trees; LGBM, LightGBM; XGB, XGBoost; SVM, support vector regression; kNN, k-nearest neighbors) was chosen. No combination was chosen consistently, which shows that the inner search does not converge on a single architecture.

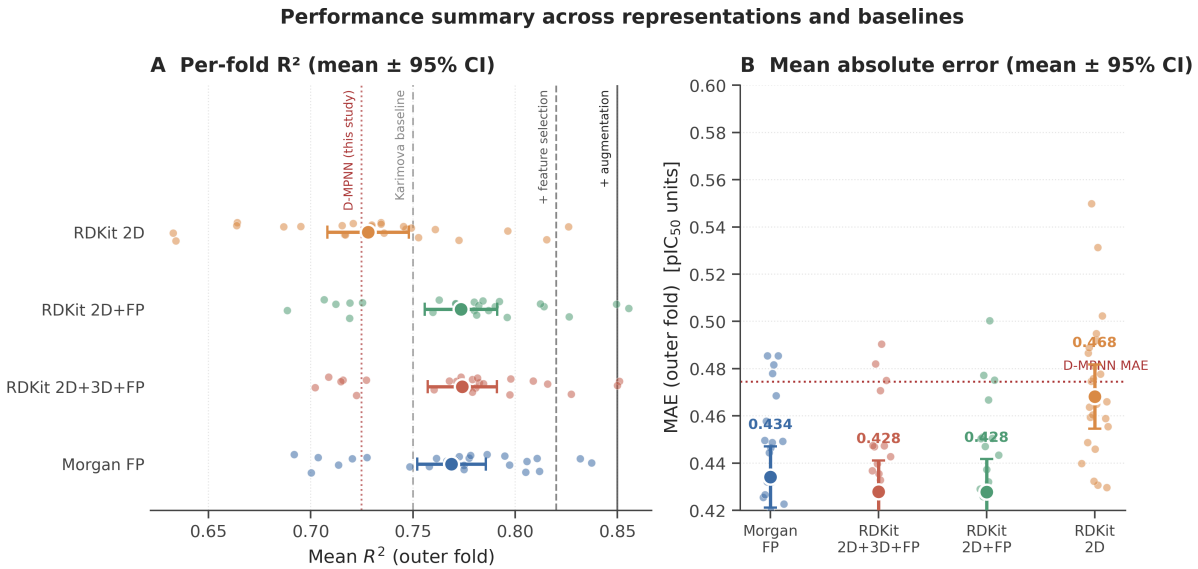


Figure S2: Per-fold performance of the four representations over the 25 outer folds. (A) R^2 of each fold (faint points), with the mean and its 95% confidence interval; (B) the same for the mean absolute error. The vertical lines mark the D-MPNN of this study and the three values reported by the original study, which come from a single hold-out split of a pipeline with the leaks described in the manuscript and are therefore shown for reference only, not as a paired comparison.

Virtual screening, docking and ADMET

Table S1: Top 10 FDA-approved candidates predicted by the classical ML model, ranked by predicted pIC_{50} and evaluated across standard ligand efficiency metrics. LE, ligand efficiency; BEI, binding efficiency index; LLE, lipophilic ligand efficiency; SEI, surface efficiency index.

Rank	Compound Name	Predicted pIC_{50}	LE	BEI	LLE	SEI
1	Trimetrexate Glucuronate	7.9272	0.40	21.46	5.19	6.74
2	Methotrexate	7.4448	0.31	16.38	7.18	3.54
3	Triamterene	7.0081	0.51	27.67	6.17	5.41
4	Pralatrexate	7.0034	0.27	14.67	6.02	3.38
5	Pyrimethamine	6.5705	0.53	26.42	4.05	8.44
6	Folic Acid	6.3379	0.27	14.36	6.38	2.97
7	Naldemedine Tosylate	6.0892	0.20	10.67	2.61	4.31
8	Bendamustine Hydrochloride	6.0850	0.36	16.98	2.82	10.43
9	Levomefolate Calcium	6.0732	0.25	13.22	6.33	3.00
10	Letermovir	5.9681	0.20	10.42	0.26	7.67

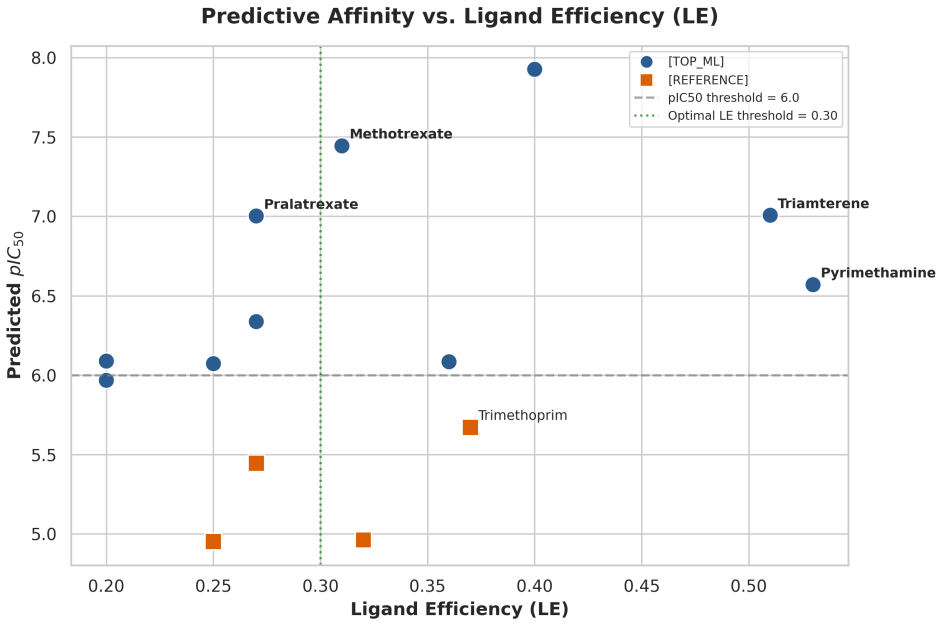


Figure S3: Predicted pIC_{50} versus ligand efficiency for the candidates advanced to docking. Blue: the ten top-ranked compounds by predicted potency; orange: reference compounds from the original study. The dashed line marks $pIC_{50} = 6.0$ and the dotted line LE = 0.30.

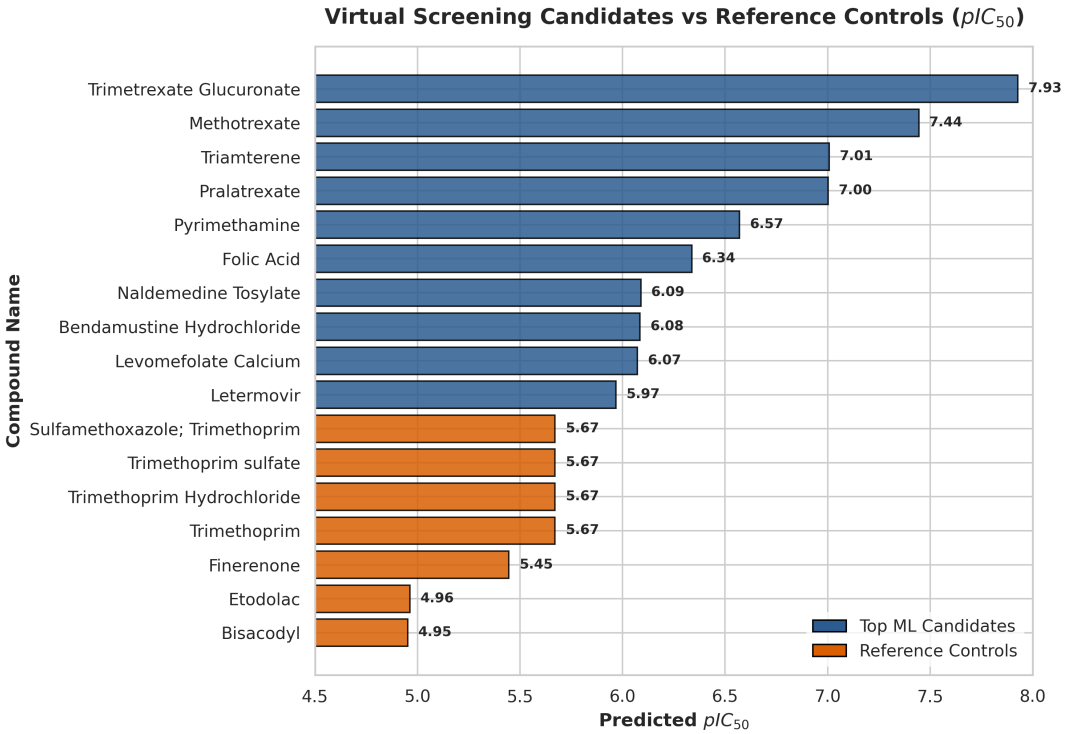


Figure S4: Predicted pIC_{50} of the screening candidates next to the reference controls of the original study. Compounds that belong to the training partition are fitted values and not predictions on unseen molecules.

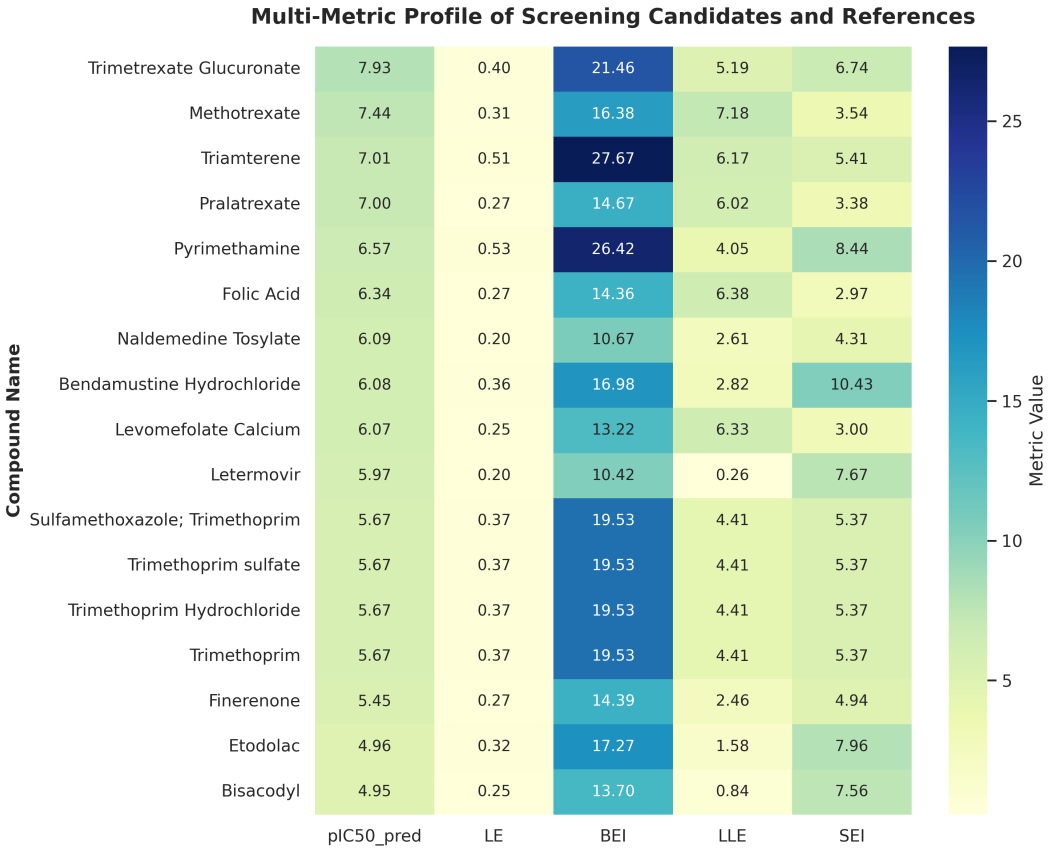


Figure S5: Efficiency metrics of the screening candidates and the reference controls. Each metric is scaled across compounds, so the colour shows a compound's standing within this set and not an absolute quality.

Table S2: Predicted ADMET profile of the 4 consensus hits and of the known inhibitors pyrimethamine and trimethoprim. Structures that appear under several entries are listed once. hERG prob. is the predicted probability of channel blockade (flag above 0.5) and Caco-2 $\log P_{app}$ the predicted permeability (oral if above -5.15). The predictions come from random forests trained on public data whose accuracy was not evaluated here, and some of these compounds are in their training sets, so the profile is a coarse prioritization and not a safety assessment.

Compound	**Pred. pIC$_{50}$**	**ΔG (kcal/mol)**	**hERG prob.**	**Caco-2 $\log P_{app}$**	**Profile**
Trimetrexate Glucuronate	7.93	−8.76	0.40	−4.86	Oral
Pralatrexate	7.00	−8.68	0.31	−5.41	Parentera
Methotrexate	7.44	−8.66	0.40	−5.87	Parentera
Triamterene	7.01	−8.07	0.80	−4.66	hERG flag
Pyrimethamine	6.57	−7.62	0.12	−4.42	Oral
Trimethoprim	5.67	−7.56	0.14	−4.77	Oral

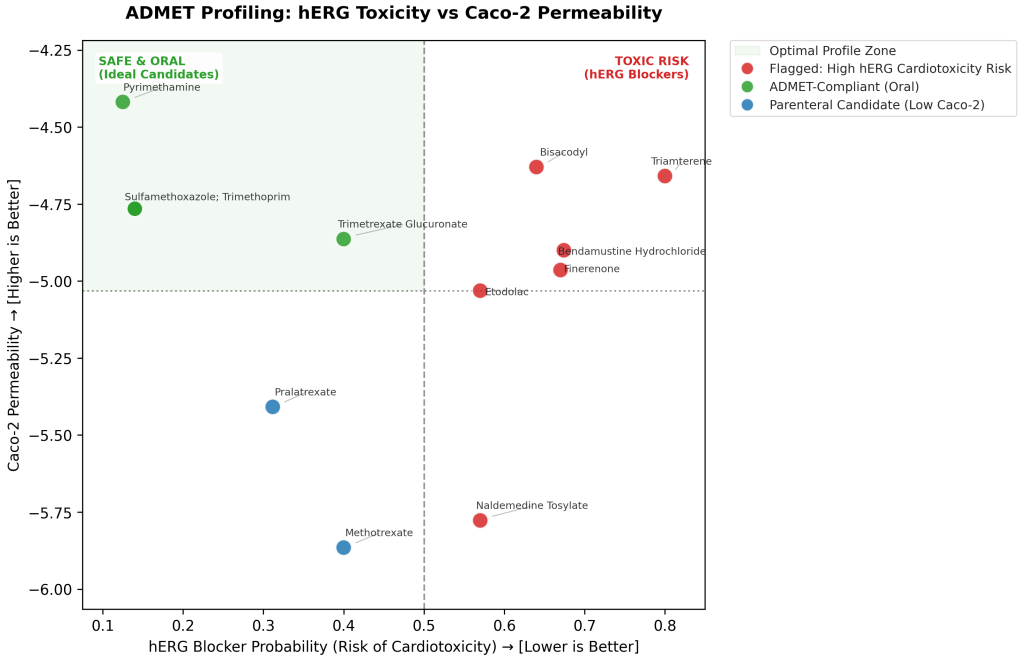


Figure S6: Predicted hERG blocking probability versus predicted Caco-2 permeability for the docked candidates that passed the Lipinski criteria. The dashed lines mark the hERG threshold and the Caco-2 threshold; the shaded quadrant combines low hERG risk with permeability classified as oral.